



If $a+b=5$ and $ab=6$, find $a^3 + b^3$.

Greedy Guarantee

Effective Learning From Teacher



Because

$$a^3 + b^3 = (a + b)^3 - 3(a + b)ab$$

Minimal Intervention

Retain Diversity From Base Model



First we solve $x^2 - 5x + 6 = 0$, Then ...

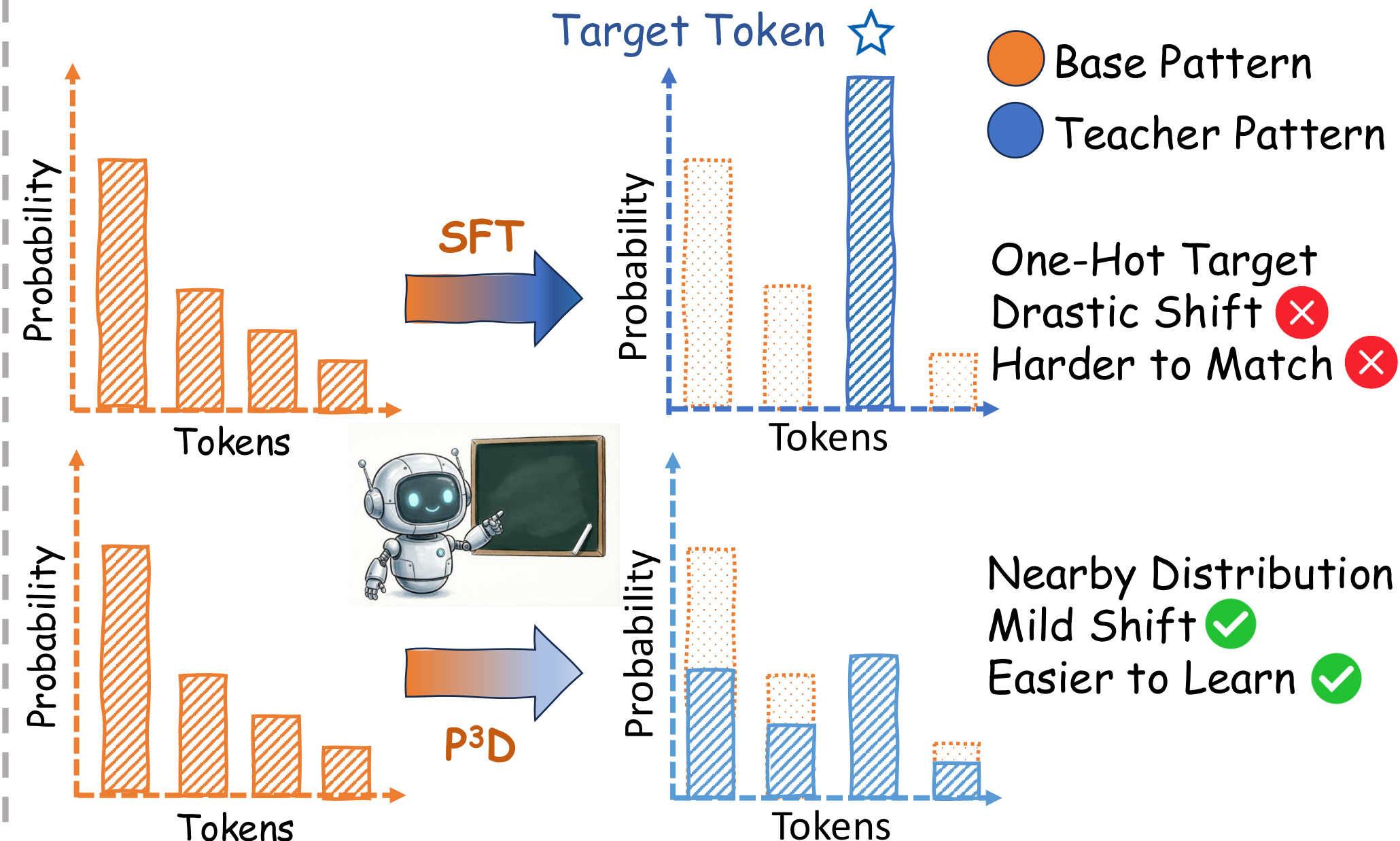
Explore Both Distributions to Forge Own Reasoning in RL



First apply

$$a^3 + b^3 = (a + b)^3 - 3(a + b)ab$$

Then



Base

After P³D

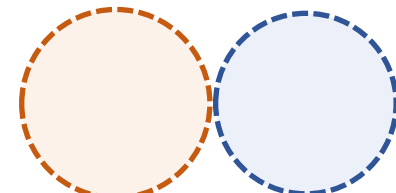
After RL



Encodes Rich Generative Priors
Lacks Problem-Solving Experience



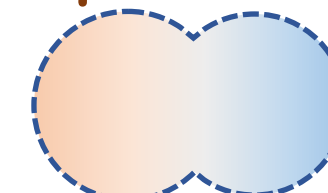
Adopt Expert Strategies
Lack Unified Thinking Patterns



No Unified Pattern



Fuse Dual Reasoning Strategies
Form Unique Thinking Patterns



Fused Unique Pattern